

Dr. Kristina P. Sinaga

PhD in Applied Mathematics | Pattern Recognition & Machine Learning Specialist

✉ kristinasinaga41@gmail.com

🌐 Kristinap09

🌐 kristinap09.github.io

🎓 Google Scholar

3,200+

Citations

7

h-index

15+

Publications

7+

Years Experience

“ The best way to predict the future is to invent it. ”

—Personal Philosophy

RESEARCH EXPERTISE & KEYWORDS

Machine Learning

Federated Learning

Multi-View Clustering

Data Science

Applied Mathematics

Statistical Modeling

Algorithm Development

Privacy-Preserving AI

Dimensionality Reduction

Unsupervised Learning

Data Integration

Computational Mathematics

Research Leadership

Academic Publishing

CORE COMPETENCIES

- › Machine Learning
- › Data Science
- › Federated Learning
- › Multi-View Clustering
- › Algorithm Development
- › Statistical Analysis
- › Dimensionality Reduction
- › Python, R, MATLAB
- › TensorFlow, PyTorch
- › Cloud Computing
- › Docker, Git
- › Privacy-Preserving AI
- › Research Leadership
- › Academic Publishing

RESEARCH INTERESTS & SPECIALIZATIONS



Clustering Methodologies: Developing innovative k-means and fuzzy c-means algorithms for single and multi-view data using novel mathematical formulations.



Pattern Recognition: Employing clustering-based dimensionality reduction techniques for feature selection and optimization in high-dimensional data analysis.



Federated Learning: Adapting conventional ML algorithms for privacy-preserving multi-client environments with enhanced data security.

PROFESSIONAL SUMMARY

Results-driven Applied Mathematics Ph.D. and Postdoctoral Researcher with **7+ years of experience** in **machine learning, data science, multi-view clustering**, and **federated learning**. Expert in **privacy-preserving machine learning, algorithm development**, and **statistical modeling**.

Research Excellence: Published in top-tier journals (IEEE TPAMI, Elsevier) with **3,000+ citations**.

Teaching Impact: Delivered courses to 200+ students with 35% engagement increase. **Editorial Service:** Peer reviewer and guest editor for IEEE, MDPI, Springer, Elsevier journals.

EDUCATION

Ph.D. in Applied Mathematics

2016 - 2020

Chung Yuan Christian University, Taiwan

- › **Dissertation:** “Multi-view Fuzzy Clustering Algorithms For Multi-View Data (多視圖數據模糊聚類演算法)”
- › **Research Focus:** Pattern recognition, multi-view clustering, dimensionality reduction
- › **Academic Excellence:** Graduated with Honors | 6 journal articles | 2,800+ citations
- › **Recognition:** Featured in IEEE Access Popular Documents

Master of Science in Mathematics

2013 - 2015

University of Sumatera Utara, Indonesia

- › **Thesis:** “Stochastic Optimization Models for Emergency Service Location Problems”
- › **Research Area:** Operations research, optimization theory, mathematical modeling

Bachelor of Science in Mathematics

2008 - 2013

University of Sumatera Utara, Indonesia

- › **Academic Distinction:** Graduated with high honors

- › **Foundation:** Analysis, algebra, and applied mathematics

ACADEMIC & RESEARCH EXPERIENCE

Independent Researcher

Oct 2025 - Present

Not Affiliated

- › **Ongoing Research:** Advanced machine learning and federated learning frameworks for multi-view data
- › **Publication Preparation:** Manuscripts for high-impact journals in preparation
- › **Guest Editor:** MDPI Future Internet - Clustered Federated Learning for Networks
- › **Peer Review:** IEEE Access, MDPI Journals, Springer, Elsevier

Postdoctoral Researcher

Oct 2024 - Sep 2025

ISTI-CNR, Italy

- › **Algorithm Development:** Novel clustering algorithms achieving 20-30% accuracy improvement
- › **Publications:** 3+ high-impact papers in federated unsupervised ML (in progress/review)
- › **Focus:** Privacy-preserving machine learning frameworks for heterogeneous data

Independent Researcher

Apr 2024 - Sep 2024

Self-Directed Research

- › **Research Autonomy:** Independent ML and clustering algorithm research
- › **Achievement:** Secured arXiv endorsement and published preprint
- › **Peer Review:** IEEE Access, Elsevier Information Fusion

Postdoctoral Fellow

Mar 2023 - Mar 2024

Chung Yuan Christian University, Taiwan

- › **Breakthrough:** Novel multi-view clustering algorithms published in IEEE TPAMI
- › **Methodology:** New algorithms for MVKM and MVFCM in federated environments
- › **Implementation:** Comprehensive MATLAB code for multi-client data processing

Professional Development

Apr 2022 - Feb 2023

Self-Directed

- › **Certifications:** ML Specialization & TensorFlow Developer (Coursera/Stanford/DeepLearning.AI)
- › **Skills:** TensorFlow, PyTorch, advanced ML methodologies

Lecturer Specialist S3 (Assistant Professor)

Nov 2020 - Mar 2022

Bina Nusantara University, Indonesia

- › **Teaching Excellence:** Mathematics & Data Analysis to 200+ students | 35% engagement increase
- › **Supervision:** 10+ student research projects
- › **Innovation:** Project-based learning improving critical thinking by 40%

PhD Research Student

Sep 2016 - Jun 2020

Chung Yuan Christian University, Taiwan

- › **Research:** Multi-view fuzzy clustering algorithms | 6 journals, 4 conferences | 2,800+ citations
- › **Impact:** h-index of 7 | International conference presentations

TEACHING EXPERIENCE

Teaching Impact Summary

180+ Students
Total Taught

80+ Undergraduate
Students

100+ Graduate
Students

4 Int'l Events
Moderated

Graduate Teaching

2020 - 2022

Bina Nusantara University

- › **Business Intelligence & Analytics:** Regular and online programs for 100+ graduate students
- › **Focus:** Data-driven decision making, statistical analysis, BI tools

Undergraduate Teaching

2021 - 2022

Bina Nusantara University

- › **Courses:** Calculus I, Discrete Mathematics for 80+ CS students
- › **Approach:** Collaborative learning and practical problem-solving

TECHNICAL SKILLS

⌘ Programming

Python

R

MATLAB

Julia

C++

SQL

LaTeX

🔬 Research Specializations

Federated Learning

Multi-View Clustering

Privacy-Preserving ML

Optimization

🧠 ML & AI Frameworks

TensorFlow

PyTorch

scikit-learn

Keras

NumPy

Pandas

☁️ Cloud & DevOps

AWS

GCP

Docker

Git

Linux

Kubernetes

OPEN SOURCE SOFTWARE

⌘ PyPI Packages

mvkm-ed 1.1.0

—PyPI | GitHub

Federated Multi-View K-Means Clustering with Rectified Gaussian Kernels

gcomvkm 0.1.0

—PyPI | GitHub

Globally Collaborative Multi-View k-Means with Entropy-Regularized Feature Reduction

SELECTED PUBLICATIONS

Publication Metrics

3,200+ Citations

|

h-index: 7

|

15+ Journal Articles

|

4 Conference Papers

Federated Multi-View K-Means Clustering

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025

✔ IF: 24.3 | Tier 1 Journal

Unsupervised k-means clustering algorithm

IEEE Access, Vol. 8, pp. 80716-80727, 2020

✔ 2,800+ Citations | IEEE Access Popular Documents

A feature-reduction multi-view k-means clustering algorithm

IEEE Access, Vol. 7, pp. 114472-114486, 2019

✓ 140+ Citations | Novel Dimensionality Reduction

Collaborative feature-weighted multi-view fuzzy c-means clustering

Pattern Recognition, Elsevier, 2021

✓ 80+ Citations | Advanced Soft Clustering

[View Complete Publication List on Google Scholar](#)

CERTIFICATIONS

Machine Learning Specialization

Stanford/DeepLearning.AI • Oct 2022

[Verify](#)

TensorFlow Developer

DeepLearning.AI • Nov 2022

[Verify](#)

PEER REVIEW & EDITORIAL SERVICE

Guest Editor

MDPI Future Internet — Clustered Federated Learning for Networks

Peer Reviewer (50+ reviews)

- IEEE Access
- IEEE TPAMI
- ACM TKDD
- Elsevier Info. Fusion
- Elsevier Applied Soft Comp.
- Pattern Recognition
- MDPI Mathematics
- MDPI Algorithms
- MDPI Sensors
- Springer Supercomputing
- Computer Methods Biomed.
- Multiple MDPI Journals

 **Recognition:** MDPI Mathematics Exceptional Reviewers List 2025

AWARDS & RECOGNITION

MDPI Exceptional Reviewer 2025

Top peer reviewers worldwide

Phi Tau Phi Honor Society 2020

Academic excellence, Taiwan

IEEE Access Popular Document

Most-read documents recognition

JST/JASSO Scholarships

Japan research exchange 2017-18

LANGUAGES

English
Advanced (C1)

Bahasa Indonesia
Native (C2)

Mandarin
Basic (A1)

Last Updated: December 31, 2025 | Academic-ATS-Optimized